

File Organizer Tool

Project Code: FO-2025-001

Version: 1.0

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1. EXECUTIVE SUMMARY

This document outlines the requirements for developing a File Organizer Tool using Python. The tool will automate the process of organizing files within a specified directory by categorizing them into appropriate folders based on file types and provide statistical reports upon completion.

2. PROJECT OVERVIEW

2.1 Purpose

To develop an intelligent file organization utility that improves directory structure and enhances file accessibility by automatically sorting files into categorized folders.

2.2 Scope

The File Organizer Tool will:

- Accept a folder path as input from the user
- Scan all files within the specified directory
- Create category-based folders automatically
- Move files to appropriate category folders
- Generate and display comprehensive statistics upon completion

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2.3 Target Users

- Corporate employees managing document repositories
- IT administrators maintaining file servers
- Individual users organizing personal directories
- Data management teams handling large file collections

3. FUNCTIONAL REQUIREMENTS

3.1 User Input Module

Requirement ID: FR-001

Priority: High

Description:

The system shall accept a valid folder path from the user.

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Acceptance Criteria:

- User can input folder path via command-line interface
- System validates the folder path existence
- System handles invalid paths with appropriate error messages
- System supports both absolute and relative paths

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3.2 File Scanning Module

Requirement ID: FR-002

Priority: High

Description:

The system shall scan all files in the specified directory.

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Acceptance Criteria:

- System identifies all files in the root directory
- System distinguishes between files and subdirectories
- System captures file metadata (name, extension, size)
- System handles files without extensions appropriately

3.3 Category Folder Creation

Requirement ID: FR-003

Priority: High

Description:

The system shall automatically create category-based folders.

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Acceptance Criteria:

- System creates folders only for categories that have files
- Folder names are clear and descriptive
- System prevents duplicate folder creation
- System handles folder creation errors gracefully

Standard Categories:

- Documents: .pdf, .doc, .docx, .txt, .rtf, .odt, .xls, .xlsx, .ppt, .pptx
- Images: .jpg, .jpeg, .png, .gif, .bmp, .svg, .ico, .webp
- Videos: .mp4, .avi, .mkv, .mov, .wmv, .flv, .webm
- Audio: .mp3, .wav, .flac, .aac, .ogg, .wma, .m4a
- Archives: .zip, .rar, .7z, .tar, .gz, .bz2
- Code: .py, .java, .cpp, .c, .js, .html, .css, .php, .sql, .json, .xml
- Executables: .exe, .msi, .apk, .deb, .dmg

- Others: All remaining file types

3.4 File Organization Module

Requirement ID: FR-004

Priority: High

Description:

The system shall move files to their appropriate category folders.

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Acceptance Criteria:

- Files are moved based on their extensions
- Original files are preserved (moved, not copied)
- System handles file name conflicts appropriately
- System maintains file integrity during movement
- System skips already-organized files if run multiple times

3.5 Statistics Generation Module

Requirement ID: FR-005

Priority: High

Description:

The system shall generate and display comprehensive statistics upon completion.

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Acceptance Criteria:

- Display total number of files processed
- Show breakdown of files by category
- Display total size organized (in MB/GB)
- Show number of folders created
- Display execution time

- Present statistics in a clear, formatted manner

4. NON-FUNCTIONAL REQUIREMENTS

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4.1 Performance

Requirement ID: NFR-001

Priority: Medium

Description:

- System shall process up to 1000 files within 60 seconds
- System shall handle files up to 2GB in size

4.2 Reliability

Requirement ID: NFR-002

Priority: High

Description:

- System shall maintain 99% accuracy in file categorization
- System shall not corrupt or lose files during organization
- System shall implement error handling for all file operations

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4.3 Usability

Requirement ID: NFR-003

Priority: Medium

Description:

- Interface shall be intuitive and require minimal user training
- Error messages shall be clear and actionable
- Statistics shall be presented in human-readable format

4.4 Maintainability

Requirement ID: NFR-004

Priority: Medium

Description:

- Code shall follow PEP 8 Python style guidelines
- Functions shall be modular and reusable
- Configuration for file categories shall be easily modifiable

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4.5 Portability

Requirement ID: NFR-005

Priority: Medium

Description:

- System shall be compatible with Windows, macOS, and Linux
- System shall use platform-independent path handling
- Python 3.8 or higher required

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5. TECHNICAL SPECIFICATIONS

5.1 Technology Stack

- Programming Language: Python 3.8+
- Standard Libraries: os, shutil, pathlib, datetime, collections
- Development Environment: Any Python-compatible IDE

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6. USER INTERFACE REQUIREMENTS

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6.1 Command-Line Interface

Welcome to File Organizer Tool v1.0

=====

Enter folder path to organize: /path/to/folder

Scanning directory...

Found 156 files to organize.

Organizing files...

[████████████████████] 100% Complete

Organization Summary

=====

Total Files Processed: 156

Total Size Organized: 2.4 GB

Execution Time: 3.2 seconds

Files by Category:

Documents: 45 files (680 MB)

Images: 67 files (1.2 GB)

Videos: 12 files (450 MB)

Audio: 18 files (120 MB)

Archives: 8 files (95 MB)

Code: 6 files (2.5 MB)

Folders Created: 6

Files Already Organized: 0

Errors: 0

Organization complete! Your files are now organized.

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7. ERROR HANDLING REQUIREMENTS

7.1 Error Scenarios

1. Invalid Path: Display error message and prompt for valid path
2. Permission Denied: Notify user of access issues
3. Disk Space Full: Alert user and halt operation
4. File In Use: Skip file and log in error report
5. Duplicate Filenames: Append timestamp or counter to filename

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8. DELIVERABLES

1. Fully functional Python application
2. Source code with inline documentation
3. User manual / README file

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9. ASSUMPTIONS AND CONSTRAINTS

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9.1 Assumptions

- Users have Python 3.8+ installed
- Users have read/write permissions for target directories
- Files are not currently in use during organization

9.2 Constraints

- No GUI implementation in initial version
- No undo functionality in version 1.0
- No cloud storage integration

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